BaM-CIM: A High Throughput Booth Algorithm-Based In-MRAM Computing Macro Using Hybrid VGSOT-MTJ/GAA-CNTFET

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***Abstract*—As artificial intelligence (AI) and computational models grow in scale, the demand for computational power and storage has significantly increased. The computing-in-memory (CIM) architecture addresses this challenge by performing computations directly within the memory array, reducing data transfer between the processor and memory. This paper introduces a Booth algorithm-based In-MRAM computing architecture (BaM-CIM) using a hybrid voltage-gated spin-orbit torque MTJ (VGSOT-MTJ) and gate-all-around carbon nanotube field-effect transistors (GAA-CNTFETs) for efficient multiply-and-accumulate (MAC) computing. The key contributions of BaM-CIM are as follows: 1) A Voltage divider reference (VDR) cell is proposed, which enables read operations using only a 2T1M cell structure. Compared to complementary read cells, the VDR reduces the area by half and achieves robust data sensing without requiring precharge/discharge operations. 2) The BaM-CIM circuit is proposed to complete 8b-W/8b-IN/21b-OUT computations in only two cycles (1.6 ns), reducing the number of cycles by 75% compared to single-bit input serial operations and by 50% compared to two-bit serial operations. 3) A three-input 8b Booth computing adder (BCA), along with Modified computing shift adder (MCSA) and Modified computing post adder (MCPA), which can achieve higher energy efficiency. BaM-CIM with 128 Kb is simulated, achieving throughput and energy efficiency of 0.93 TOPS and 258.4 TOPS/W, respectively, at a 0.6 V supply voltage and 1.28 TOPS and 169.5 TOPS/W, respectively, at a 0.8 V supply voltage with 8b-IN, 8b-W, and 21b-OUT.**

***Index Terms*—Computing-in-memory, MRAM, Booth algorithm, adders, GAA-CNTFET, VGSOT-MTJ.**

I. Introduction

With the rapid development of artificial intelligence (AI) and the increasing scale of computational models, there has been a significant rise in the demand for computational power and storage [1]. As transistor sizes shrink, the processing capabilities of integrated circuits improve, making logic gates faster, and the processing speed of CPUs far exceeds the read-write speed of memory. This leads to increased waiting times for the CPU, reducing computational efficiency. Furthermore, with the growth of data volumes, frequent memory accesses also result in high energy consumption, becoming a major limiting factor for large-scale data processing performance. The computing-in-memory (CIM) architecture, which performs computations directly within the memory array, reduces the data transfer between the processor and memory [2], [3], [4], [5], [6]. This approach holds promise for alleviating the memory wall bottleneck, enhancing computational speed, and lowering energy costs.

In neural networks (NN), the weights post-training remain almost unchanged, and there is no need for frequent refresh operations during the inference process [7], [8], [9], [10]. Spintronic devices, with their excellent electrical properties [11], [12], [13], enable the realization of ultra-low power non-volatile memory [14], [15]. This feature gives MRAM an advantage in hardware design for NNs. Moreover, MRAM exhibits excellent low-power characteristics for data writing, making it an attractive option for designing CIM architectures [16], [17], [18], [19].

The NN computation process involves both multiplication and accumulation computing (MAC) operations and nonlinear activation operations, with the former accounting for the vast majority of computations. To achieve high-precision MAC operations, current design approaches typically use bit-serial inputs and multi-cycle computations to perform partial bit-width MAC operations for weights and input data. These partial results are then accumulated using bit-shift operations according to the bit width of the weights [1], [18], [20], [21]. However, despite these advancements, current MRAM-based CIM architectures still face several challenges:

Optimization of readout methods:Unlike the storage mode, where sense amplifiers (SA) are solely used to read data, MRAM in computational mode seeks to achieve faster read speeds to match high-speed operations, or utilizes embedded computing units with data reading capabilities. This evolution has led to the development of alternative data readout methods,

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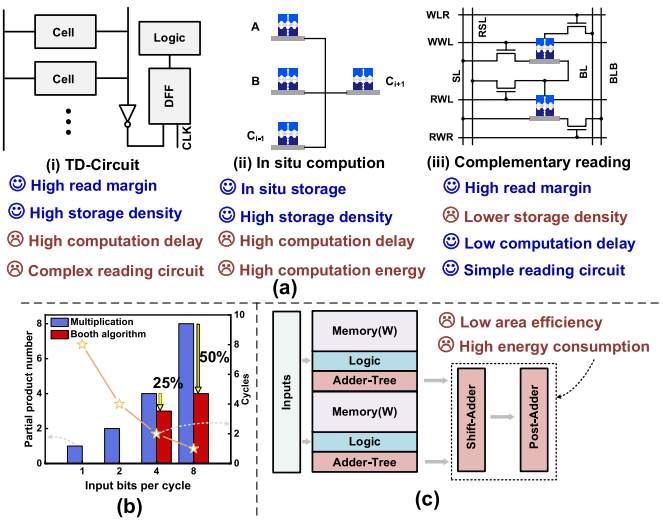


Fig. 1. (a) Three different ways of readout methods. (b) The number of operation cycles and the number of partial products varies with input bits per cycle in bit-serial computing mode. (c) Conventional CIM architecture for MAC operations.

as shown in **Fig. 1(a)**. In [22] and [23], a time domain CIM (TD-CIM) scheme was proposed. While this scheme can improve the read margin, it requires additional D flip-flops for sampling, which requires accurate clock sampling. In [24], a circuit was designed to complete logic AND and full-adder operations within the array and store the results in situ. However, this method involves multiple steps to execute the full-adder operation and also introduces additional energy consumption and delays during the MTJ write process. Meanwhile, [25] proposed a complementary unit-based design for voltage division that also participates in computation. This approach eliminates read delays but requires more transistors for storage, which results in reduced storage density.

Trade-off in bit-serial computation:Bit-serial computation often involves a trade-off between the number of operation cycles and the complexity of the computation circuit. A lower bit-serial input bit-width per cycle reduces the complexity of the bit-wise operation circuit at the cost of longer operation cycles [8], [26], [27]. Conversely, when the input bit-width per cycle increases, the circuit complexity grows, but the operation cycle becomes shorter [18], [28], as shown in **Fig. 1(b)**. The Booth algorithm reduces the number of partial products by exploiting consecutive bits of the multiplier, thereby reducing the multiplication steps while maintaining a short operation cycle. This reduces the circuit complexity and improves the efficiency of the MAC operation. When a Radix-4 Booth multiplier is used, the number of partial products is reduced by 25% for multiplying two 4-bit numbers, and by 50% for multiplying two 8-bit numbers. Ref. [28] compared to traditional single-bit input serial CIMs, the proposed scheme, by using the Booth algorithm, reduces the number of cycles and bit-wise multiplications by nearly 50%, resulting in a 17.3% reduction in energy consumption.

Overhead of adders in existing CIM systems: In peripheral computation circuits, Adder trees are used to sum the results of multiplications performed by the overall array, shift adders are used to accumulate results with different bit-width weights, and post adders are used to shift and accumulate computation results from different cycles (from Input LSB to MSB), as shown in **Fig. 1(c)** [2], [25], [29], [30]. These three types of adders are essential for completing high-precision MAC operations. The area and energy consumption occupied by adders in the entire circuit are also significant, and thus, optimizing adders is crucial for improving circuit performance. In [18], a bit-level in-memory Booth multiplication architecture was adopted to optimize in-memory MAC computation. Compared to traditional single-bit input serial architectures, this design reduces the computation cycles and bit-level multiplication operations by nearly 50%. However, this work did not optimize the used adder trees, leaving room for further improvements in circuit energy efficiency [31], [32].

Therefore, to overcome the aforementioned issues, this work introduces a Booth algorithm-based in-MRAM computing architecture (BaM-CIM) using a hybrid voltage-gated spin-orbit torque MTJ (VGSOT-MTJ) and gate-all-around carbon nanotube field-effect transistors (GAA-CNTFETs) for efficient multiplication and addition computing. The contributions of the proposed BaM-CIM are as follows:

1. A Voltage divider reference (VDR) is proposed, which utilizes voltage division between itself and a 2T1M structure to read data. The voltage is then reshaped using an inverter to output the stored data, with a read margin of up to 115mV. This data reading method does not require bitline precharge and discharge operations, eliminating data reading delay while ensuring both storage density and read margin.
2. A multiplication circuit based on the Booth algorithm is designed. The number of partial products in the multiplication is reduced by 25%. A system-level architecture is realized, achieving 8b-W/8b-IN/21b-OUT computation in just two operation cycles (1.6ns) with a performance of 169.5 TOPS/W. The improved Booth encoder uses pass transistor logic, reducing the number of transistors by 59%.
3. A three-input 8-bit adder, the Booth computing adder (BCA), is designed based on the characteristics of partial products generated by the Booth algorithm. Leveraging optimized logic structures, the BCA reduces energy consumption by 35% compared to traditional adders. Modified computing shift adder (MCSA) and modified computing post adder (MCPA) are also designed for the subsequent shift adder and post adder stages. The MCSA achieves a 30% reduction in energy consumption, while the MCPA further improves efficiency by 38%, both while lowering the transistor count compared to conventional designs.

The remainder of this paper is organized as follows: **Section II** presents the devices used in BaM-CIM and an introduction to the Booth algorithm. **Section III** discusses the principles of VDR, Booth encoding and decoding circuits, the structures and principles of three high-performance adders: BCA, MCSA, and MCPA, as well as the overall circuit architecture. **Section IV** presents performance data of each part of the circuit, including the robustness of VDR, adder performance analysis, overall circuit performance analysis,

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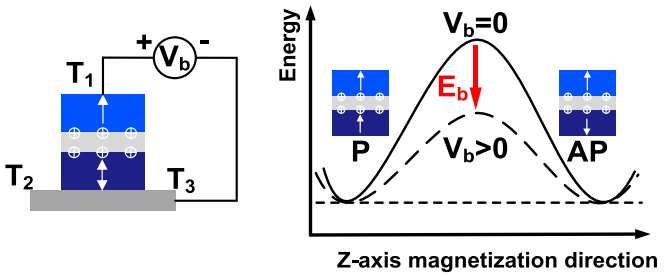


Fig. 2. Schematic of VGSOT-MTJ device.(E_b represents the energy barrier).

comparisons with other works, and layout evaluation. **Section V** provides a summary of the entire work.

II. Background

A. VGSOT-MTJ

The voltage-gated spin-orbit torque (VGSOT) effect utilizes voltage-controlled magnetic anisotropy (VCMA) to assist spin-orbit torque (SOT). Based on the VCMA effect, when a positive bias voltage is applied to the external oxide layer, the energy barrier between the parallel (P) and anti-parallel (AP) states of the magnetic tunnel junction (MTJ) is reduced, as shown in **Fig. 2**. Therefore, introducing the VCMA effect in p-MTJs with an anti-ferromagnetic/ferromagnetic/oxide structure can lower the critical switching current (I_{SOT}), improving switching reliability and reducing energy consumption during switching [33]. The VGSOT-MTJ device has three terminals: T_1, T_2, and T_3. When a positive bias voltage (V_b) is applied to T_1, the minimum write current between T_2 and T_3 is significantly reduced. Moreover, the high resistance characteristics of the VGSOT-MTJ device help increase the voltage margin, improving computational accuracy. At the same time, when the bitline is activated, it reduces static current, lowering energy consumption [34].

B. GAA-CNTFET

With the continuous development of Moore's Law, the size of silicon-based transistors has been reduced to the nanoscale, approaching atomic dimensions. This leads to issues such as quantum tunneling, which can affect device reliability and performance. Consequently, exploring new materials, devices, and design methods has become an inevitable challenge. Gate-all-around carbon nanotube field-effect transistors (GAA-CNTFETs) are field-effect transistors that use carbon nanotubes (CNTs) as the conductive channel. CNTs offer excellent electronic properties, including high carrier mobility, good thermal stability, and mechanical strength. **Fig. 3(a)** shows a schematic of the GAA-CNTFET model. The carbon nanotubes are designed to surround the gate structure, enhancing gate control, improving the I_{ON}/I_{OFF} ratio, and further reducing leakage current [35], [36]. **Fig. 3(b)** presents the voltage transmission characteristics of inverters built with 14nm-FinFETs and GAA-CNTFETs. Both 14nm-FinFET and GAA-CNTFET employ the minimum feature size, with the specific parameters of GAA-CNTFET provided in **Table V**.

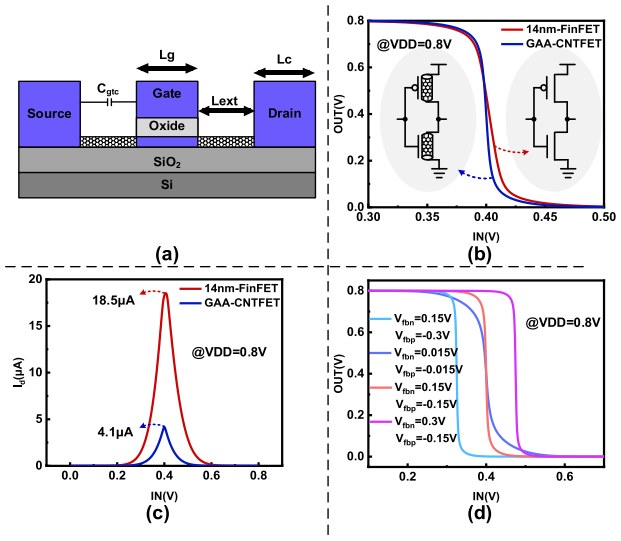


Fig. 3. (a) Schematic of GAA-CNTFET. (b) Voltage transmission characteristics of inverter composed of 14nm-FinFET and GAA-CNTFET. (c) I-V characteristic curves of two inverters. (d) Voltage transmission characteristics of inverter composed of CNTFET with different flat-band voltages.

Fig. 3(c) shows the current flowing through the power supply during switching for both types of inverters, where the inverter made from CNTFETs demonstrates lower power consumption. CNTFETs can also adjust the flat-band voltage (V_{fb}) of the transistor to enhance inverter gain and modify the inverter's threshold voltage, providing greater flexibility in circuit design, as shown in **Fig. 3(d)**. Based on these characteristics, CNTFETs are expected to be used in future high-speed, low-power integrated circuits, with great potential for digital integrated circuit design.

At the same time, from a process perspective, CNTFETs can be manufactured using back-end-of-the-line-compatible processes at low temperatures ($\leq 300°C$) [37], facilitating their 3D integration with MTJs in a monolithic multilayer structure to improve storage density [38]. Therefore, GAA-CNTFETs and SOT-MTJs are considered promising candidates for building future energy-efficient computing memory macros.

C. Booth Algorithm

The core idea of the Booth algorithm is to analyze the consecutive sequences of 0s and 1s in the binary representation of the multiplier to reduce unnecessary addition and subtraction operations [39]. The algorithm introduces what is called "Booth encoding," as shown in **Table I**, which combines every two bits of the multiplier to form a new encoding sequence. Booth encoding: 0 represents that the partial product is equal to 0, 1 Represents that the partial product is equal to the multiplicand (M) itself, -1 represents that the partial product is equal to the two's complement of the multiplicand, 2 represents that the partial product is equal to the multiplicand left-shifted by one bit, and -2 represents that the partial product is equal to the two's complement of the multiplicand left-shifted by one bit [31], [32], [40]. This encoding sequence then determines whether an addition or subtraction operation should be performed. A multiplier designed using the Booth algorithm is referred to as a Booth multiplier. The Booth

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TABLE I
BOOTH ENCODING TABLE

IN_{+1}	IN	IN_{-1}	Booth Encoder	Operation
0	0	0	0	0
0	0	1	+1	+M
0	1	0	+1	+M
0	1	1	+2	+2M
1	0	0	-2	-2M
1	0	1	-1	-M
1	1	0	-1	-M
1	1	1	0	0

TABLE II
EXAMPLE OF BOOTH ALGORITHM

X	000111
×	×
Y	0001010
Booth Encoder	0 1 1
Operation	
X	00000111 → C
X,Shift two bits	00011100 → B
0,Shift four bits	00000000 → A
Summation	00100011

multiplier is a mature and efficient method that improves speed and reduces power consumption by reducing the number of partial products. The number of partial products that can be reduced depends on the radix of the Booth algorithm. The higher the radix of the Booth multiplier, the more partial products can be eliminated, but the encoding and decoding parts become more complex. Therefore, choosing the appropriate Booth multiplier is crucial for improving the performance of multiplication operations [41]. In this paper, we select a radix-4 Booth multiplier to strike a balance between performance and complexity.

Table II shows the multiplication of two 4-bit binary numbers, X = 0111 and Y = 0101, using the base-4 Booth algorithm. The three partial products generated during the operation are named A, B, and C, respectively, to facilitate subsequent understanding of the internal operations of BaM-CIM.

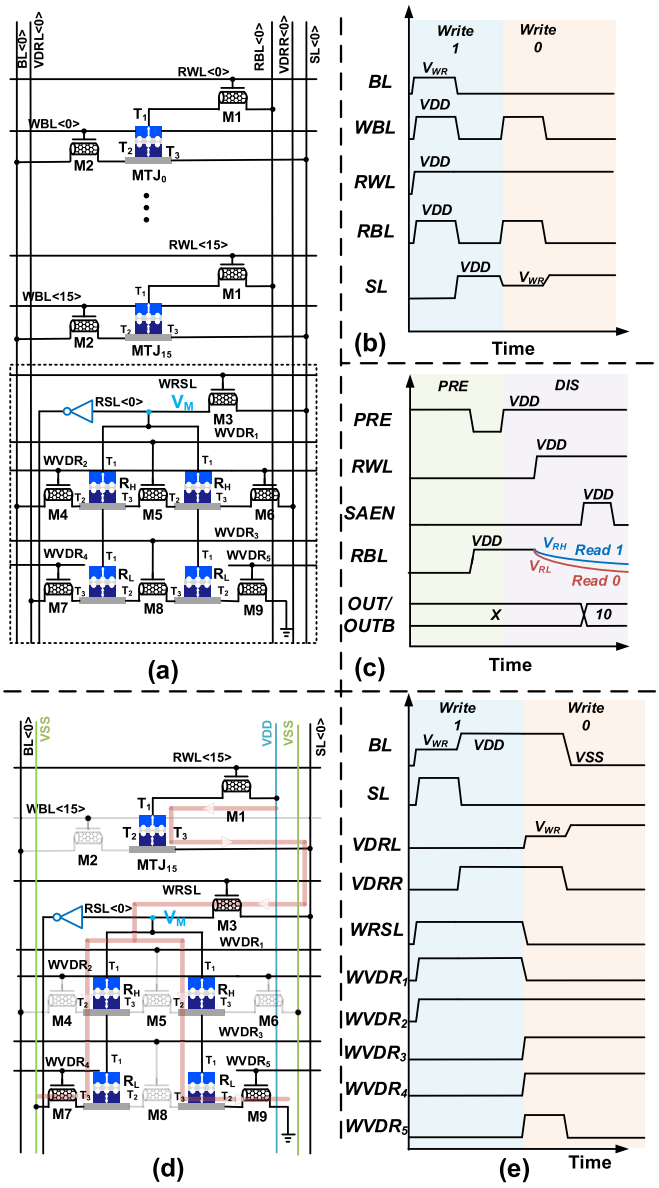


Fig. 4. (a) Schematic of 2T1M cell and VDR. (b) Write timing diagram of VGSOT-MTJ. (c) Read timing diagram of VGSOT-MTJ. (d) Computation mode of VDR. (e) Write timing diagram of VDR.

III. PROPOSED BAM-CIM FOR CIM OPERATIONS

A. The Principles of VDR

The storage unit uses the most basic 2T1M cell, as shown in **Fig. 4(a)**. Each column contains 16 cells, and at the bottom of each column, a VDR is included to assist with computations. The equivalent resistance between the bitline RSL and ground is given by $(R_H + R_L)/2$.

1) Write Operation of Storage Cell: The timing diagram for writing data "1" (R_H) is shown in **Fig. 4(b)**. The write process of the VGSOT-MTJ is divided into two stages. In the first stage, the RWL and WBL signals are set to VDD, turning on M1 and M2. The RBL signal is set to VDD, SL is VSS, and the BL voltage is set to V_{WR}. To ensure $V_b > 0$ and reduce I_{SOT} current, V_{WR} is set to 300 mV. This generates an SOT (Spin-Orbit Torque) current flowing from the T_2 terminal to the T_3 terminal. In the second stage, the RWL signal is VDD,

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WBL is VSS, M1 is turned on, and M2 is turned off. The RBL signal is set to VSS, and SL is set to VDD, realizing $V_b < 0$ and maintaining it for a certain period to complete the write operation. To write data "0" (R_L), it is required that $V_b > 0$ in the first stage, which generates an SOT current flowing from T_3 to T_2. In the second stage, $V_b < 0$ to ensure the completion of the writing process, as shown in **Fig. 4(b).**

2) Read Operation: The read mode is the same as the classical 2T1M cell. Transistors M1 and M2 are turned off, the SL signal is set to VSS, and the RBL is precharged to VDD. Then, transistor M1 is turned on, and the RBL discharges through the MTJ. After a certain period, the SAEN signal is activated, allowing the information stored in the MTJ to be read out through the SA, as shown in **Fig. 4(c)**. The blue and red lines represent the voltage changes on the corresponding RBL during the discharge of R_H and R_L, respectively. Due to the difference in discharge rates, the data can be read immediately after SAEN is enabled.

3) Write Operation of VDR: The principle is the same for the write process of the VDR. During the VDR write process, writing R_H and R_L requires SOT currents flowing in opposite directions between terminals $T_2$and T_3. The timing diagram for the write operation is shown in **Fig. 4(e)**. Taking writing to R_H as an example: turn off M7, M8, and M9, and turn on M3, M4, M5, and M6. In the first stage of writing, set the BL voltage to V_{WR}, the SL voltage to VDD, and the VDRR voltage to VSS to achieve $V_b > 0$, thereby generating an SOT current flowing from terminal T_2 to T_3. In the second stage of writing, set the voltages of BL and VDRR to VDD, and SL to VSS to achieve $V_b < 0$, ensuring the writing to R_H.

To enter computation mode, the weight W is first written into one of the 16 cells in a column. M3,M7 and M9 are turned on. The RBL voltage is set to VDD, forming a loop starting from RBL, passing through MTJ and the VDR, and eventually reaching ground, as shown in **Fig. 4(d)**.

In this work, the minimum size GAA-CNTFET has a resistance R_{ON} of 3.3 KΩ, while the VGSOT-MTJ, with a tunneling magnetoresistance (TMR) ratio of 200%, has R_L and R_H values of 333 KΩ and 974 KΩ, respectively. Therefore, to simplify the analysis, we can neglect the transistor resistance effects. According to the voltage divider formula, the voltage at the V_M node is given by:

$$V_{RSL} = V_M = \frac{R_H + R_L}{R_H + R_L + 2R_{MTJ}} VDD \tag{1}$$

when the weight is 1, the MTJ is in a high resistance state, and the value of V_M is lower:

$$V_{RSL} = V_{ML} = \frac{R_H + R_L}{3R_H + R_L} VDD \tag{2}$$

when the weight is 0, the MTJ is in a low resistance state, and the value of V_M is higher:

$$V_{RSL} = V_{MH} = \frac{R_H + R_L}{R_H + 3R_L} VDD \tag{3}$$

B. Booth Encoding

As discussed earlier, before performing multiplication using the Booth algorithm, the multiplier needs to be encoded. The input data $IN[3:0]$ is mapped to the multiplier, where adjacent 3 bits, defined as (IN_{+1}, IN, IN_{-1}) need to be encoded. Three signals, TWO, NEG, and ZERO, represent shift, negate, and zero operations, respectively. For example, if the values of (IN_{+1}, IN, IN_{-1}) are "100" the Booth encoding value is "−2", and the corresponding signals for TWO, NEG, and ZERO would be "110". Thus, the logical expressions for these signals are:

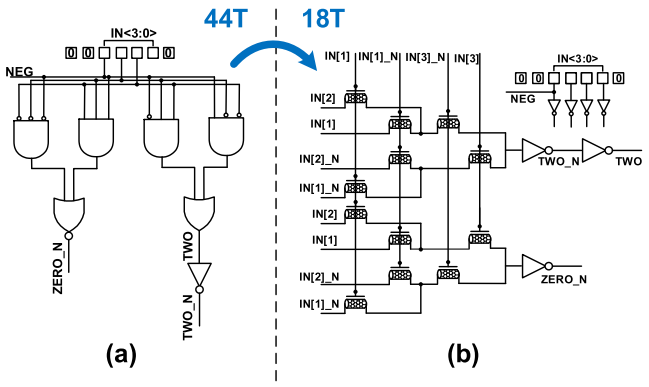


Fig. 5. (a) The conventional Booth encoder using CMOS logic (44T). (b) The proposed PTL-based Booth encoder (18T).

$$ZERO = IN_{+1}ININ_{-1} + \overline{IN}_{+1}\overline{ININ}_{-1} \tag{4}$$

$$TWO = \overline{IN}_{+1}ININ_{-1} + IN_{+1}\bar{I}N\bar{I}N_{-1} \tag{5}$$

$$NEG = IN_{+1}\overline{ZERO} \tag{6}$$

The input $IN[3:0]$ is divided into three parts: $(0, 0, IN[3])$, $(IN[3], IN[2], IN[1])$ and $(IN[1], IN[0], 0)$each corresponding to a partial product A, B, and C. For each partial product, a set of control signals, TWO, NEG, and ZERO, is generated. For example, partial product B is controlled by TWOB, NEGB, and ZEROB, while partial product C is controlled by TWOC, NEGC, and ZEROC.

There are two methods to generate the three control signal sets: The first method uses a single Booth encoder, which generates the three control signals over three cycles. The advantage of this approach is fewer encoders, but the drawback is the need for more cycles and partial product storage for each signal set. The second method uses three Booth encoders, generating the three control signals in one cycle. The advantage here is fewer cycles and no need to store partial products, as all operations (generation and summing) occur within a single cycle. However, this method requires more encoders, increasing the area and power consumption. To ensure fast computation in the circuit, this work adopts the second method. Furthermore, optimizations and improvements have been made to the encoder design.

References [18], [28], [31], and [40] use logic gate designs for the Booth encoder, as shown in **Fig. 5(a)**. This work, however, optimizes the Booth encoder by using pass transistor logic, as shown in **Fig. 5(b)**. Compared to logic gates, this approach reduces the number of transistors by 59%. Finally, inverters are used to shape the output signals, ensuring that there is no threshold loss. Moreover, since the generation of partial product A only requires an AND gate (no encoding is

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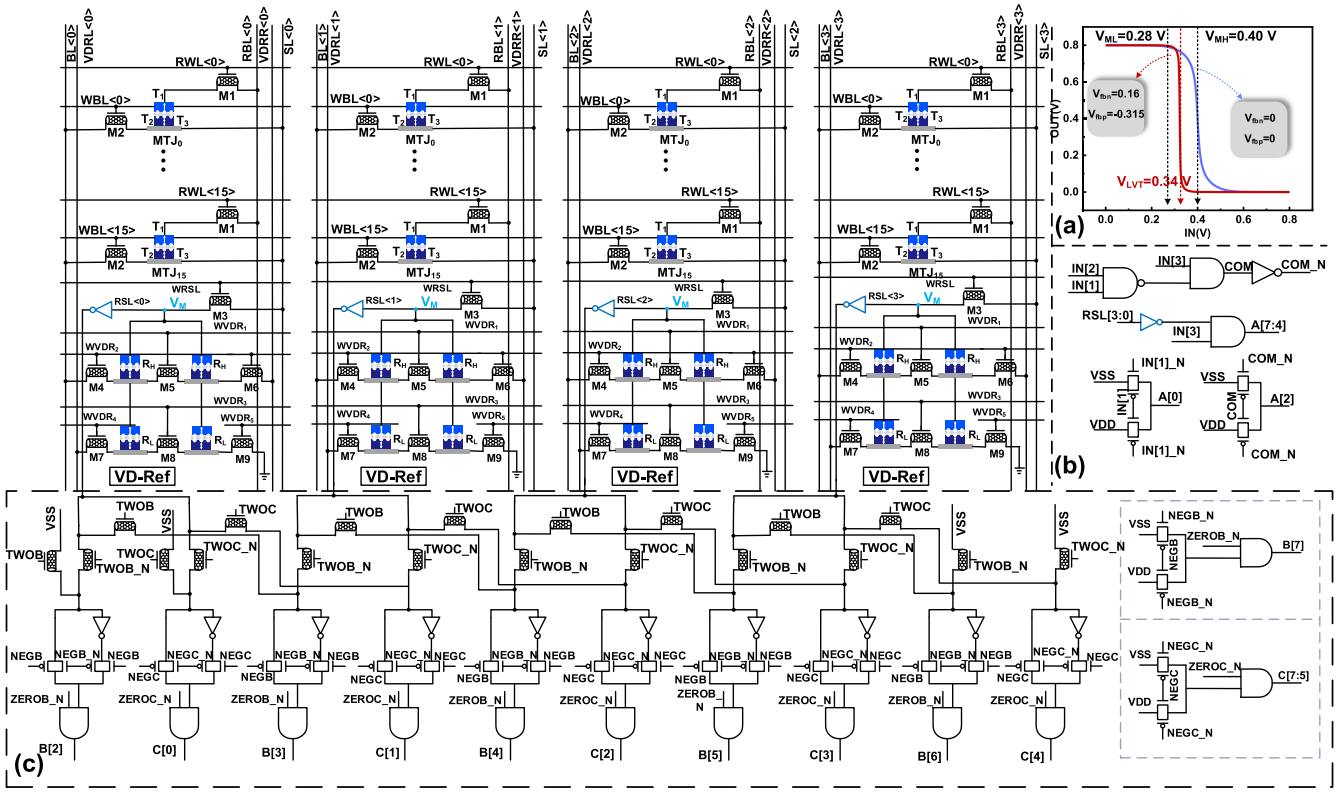


Fig. 6. Booth decoding circuit: (a) LVT inverter voltage transfer characteristics. (b) Partial Product A generation circuit. (c) Partial product B and C generation circuit.

needed), only two encoders are necessary, reducing both area and power consumption while maintaining the circuit's logic.

C. Booth Decoding

To adjust the read-out data to full swing, an inverter with a low threshold (LVT-INV) is used to shape the voltage V_M. In the LVT inverter, the flat-band voltage of the n-CNTFET is V_{fbn} = 0.16V and the flat-band voltage of the p-CNTFET is V_{fbp} = −0.315V, ensuring that the threshold voltage of the inverter is $(V_{MH} + V_{ML})/2$, as shown in **Fig. 6(a)**. This way, when V_{ML} passes through the inverter, the output signal is high, representing W as "1", and when V_{MH} passes through the inverter, the output signal is low, representing W as "0".

1) Partial Product A Generation Circuit: The circuit for generating partial product A is shown in **Fig. 6(b)**. Since the corresponding consecutive three bits of the multiplier for partial product A are $(0, 0, IN[3])$, there are only two possibilities: when $IN[3] = 1$, partial product A is equal to $W[3:0]$; when $IN[3] = 0$, partial product A is all zeros. This can be implemented using the AND logic between $IN[3]$ and $W[3:0]$.

Since partial product A is generated by the third Booth encoding, it requires a 4-bit left shift during accumulation. As the product of $IN[3:0] \times W[3:0]$ yields an 8-bit result, the lower four bits of A are zero-padded. This allows the "+1" from the two's complement conversion to be embedded into these bits when complementing partial products B and C. Here, "+1" refers to the increment in two's complement conversion, not the operation shown in **Table I**.

Partial product B is generated by the second Booth encoding and requires a left shift by two bits during subsequent accumulation. When a "+1" operation is needed for partial product B, it will be carried out at the $B[2]$ bit. We can transfer this "+1" operation to the $A[2]$ bit. A signal COM is defined to represent the complement of the partial product B. When COM = 1, $B[2]$ requires "+1", i.e., $A[2] = 1$; when COM = 0, no "+1" is needed, i.e., $A[2] = 0$. Similarly, a "+1" operation may occur at $C[0]$, and we transfer this operation to $A[0]$. Since the consecutive three bits for partial product C are $(IN[1], IN[0], 0)$, the last bit is always 0, so the $IN[1]$ signal can directly represent the complement operation for partial product C.

2) Partial Product B Generation Circuit: **Fig. 6(c)** shows the generating circuit for the partial product B and C. After V_M is shaped by the inverter, it becomes the W signal, which is sent to the Booth decoding circuit to generate partial product B. Since partial product B is generated by the second Booth encoding, it requires a left shift by two bits during subsequent accumulation, i.e., only the bits $B[7:2]$ need to be considered, with $B[1:0]$ set to 0.

The TWOB and TWOB_N signals are used to determine whether the W signal needs to be shifted. When TWOB = 0 and TWOB_N = 1, no shifting is performed, and $B[5:2]$ stores the weight $W[3:0]$ after completing the three steps (TWO, NEG, ZERO) operation. $B[7:6]$ is controlled by NEGB, determining whether to add "1" or "0". When TWOB = 1 and TWOB_N = 0, the data needs to be shifted by one bit, and $B[6:3]$ stores the weight $W[3:0]$ after

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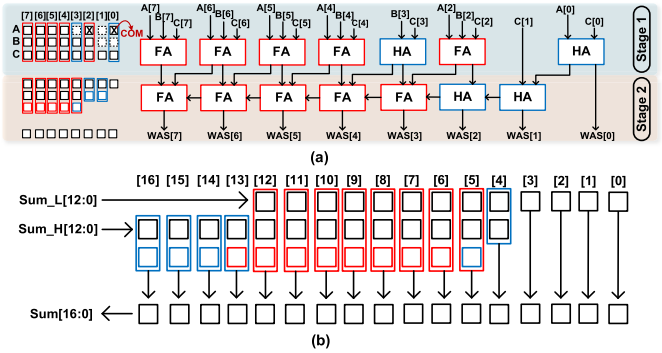


Fig. 7. (a) Schematic of BCA. (b) Schematic of MCSA.

completing the three steps. $B[2]$ is padded with 0, and $B[7]$ is determined by NEGB, either adding "1" or "0".

Finally, before generating the full partial product B, the ZEROB_N (inverted signal of ZERO) signal is checked to determine whether to take 0. When ZEROB_N = 0, $B[7:2]$ is set to 0; when ZEROB_N = 1, $B[7:2]$ is output normally.

3) Partial Product C Generation Circuit: Partial product C is generated by the first Booth encoding. During the subsequent accumulation of partial products, no left shift is required, and it is an 8-bit number, $C[7:0]$. The TWOC and TWOC_N signals are used to determine whether the W signal needs to be shifted. When TWOC = 0 and TWOC_N = 1, no shifting is performed, and $C[3:0]$ stores the weight $W[3:0]$. $C[7:4]$ is controlled by NEGC, determining whether to add "1" or "0". When TWOC = 1 and TWOC_N = 0, the data is shifted by one bit, and $C[4:1]$ stores the weight $W[3:0]$. $C[0]$ is padded with "0", and $C[7:5]$ is controlled by NEGC, either adding "1" or "0".

Finally, before generating the full partial product C, the ZEROC_N signal is checked to determine whether to take "0". When ZEROC_N = 0, $C[7:0]$ is set to 0; when ZEROC_N = 1, $C[7:0]$ is output normally.

D. Booth Computing Adder (BCA)

The next step is to sum the three partial products, A, B, and C. Based on the special structure of these three data sets and the sum compression approach of the Wallace tree, a special 3-input 8-bit adder is designed [42], named the Booth computing adder (BCA).

As shown in **Fig. 7(a)**, the red rectangles represent full adders, the blue rectangles represent half adders, the blue squares represent the carry from the lower bit half adders, and the red squares represent the carry from the lower bit full adders. The black squares represent data in the corresponding bit, and the black dashed squares represent places where no data exists (data = 0). The black squares inscribed with "X" represent the complement bit from partial products B and C.

The BCA is divided into two stages:

- Stage 1:

For bit [0], only $A[0]$ and $C[0]$ are present, so a half adder is used to sum them.

For bit [1], no operation is required in Stage 1.

For bit [3], only $B[3]$ and $C[3]$ are present, so a half adder is used to sum them.

For the remaining bits, all three partial products (A, B, and C) are involved, so full adders are used to sum them.

- Stage 2:

For bit [0], only one data is left, so it is directly output as the final result.

For bits [1] and [2], two data are present, so half adders are used.

For the remaining bits, all three data are involved, so full adders are used, and the carry from the highest bit is ignored.

Therefore, BCA completes the

$$WAS_0[7:0] = IN_0[3:0] \times W_0[3:0]. \tag{7}$$

In traditional three 8-bit number additions, an 8-bit adder and a 9-bit adder are needed, totaling 15 full adders and 2 half adders. However, the BCA only requires 10 full adders and 4 half adders, making it more efficient. The power consumption comparison between these two approaches will be discussed in **Section IV**.

Each $WAS[7:0]$ is then sent to the Adder tree for summing. $W[7:4]$ and $W[3:0]$ each have 32 sets, and each of the two Adder trees performs the following calculations.

$$Sum_L[12:0] = \sum_{i=0}^{31} IN_i[3:0] \times W_i[3:0] \tag{8}$$

$$Sum_H[12:0] = \sum_{i=0}^{31} IN_i[3:0] \times W_i[7:4] \tag{9}$$

E. Modified Computing Shift Adder (MCSA)

Next, the $Sum_L[12:0]$ and $Sum_H[12:0]$ need to be added after shifting. Unlike the traditional approach, which uses a 17-bit adder and fills with zeros during the shift and addition, we take advantage of the fixed zero positions generated during the shift and design a new shift adder. Since only 13 bits are involved in the actual computation, we name this adder the Modified computing shift adder – 13b (MCSA-13b).

As shown in the **Fig. 7(b)**, after $Sum_H[12:0]$ is left-shifted by four bits, it is added to $Sum_L[12:0]$. The first four bits of $Sum_L[12:0]$ and the last four bits of $Sum_H[12:0]$ will form fixed zero positions, so the first four bits of $Sum_L[12:0]$ do not need to participate in the computation and are directly output as the result. The bit [4] and bits [16 : 13] are summed using half adders, while the remaining bits are summed using full adders. The final result is $Sum_1[16:0]$.

$$Sum_1[16:0] = \sum_{i=0}^{31} IN_i[3:0] \times W_i[7:0] \tag{10}$$

In contrast to the traditional 17-bit shift adder, which requires 16 full adders and one half adder, the MCSA-13b only requires 8 full adders and 5 half adders. The power consumption comparison between these two approaches will be discussed in **Section IV**.

F. Modified Computing Post Adder (MCPA)

$Sum_1[16:0]$ represents the result of computing $IN[3:0] \times W[7:0]$ for one cycle. To compute the entire

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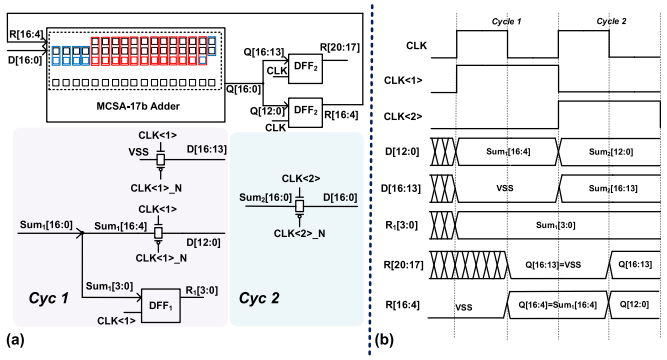


Fig. 8. (a) Schematic of MCPA. (b) Timing diagram of the MCPA.

$IN\,[7:0] \times W[7:0]$, two cycles are required, and the two generated $Sum_{1-2}[16:0]$ values are then added after shifting. This requires the use of the Modified computing post adder (MCPA).

As shown in **Fig. 8(a)**. In the first cycle, $Sum_1[3:0]$ does not participate in the computation and is stored in the register as $R_1[3:0]$ using a flip-flop. $Sum_1[16:4]$ is passed through a transmission gate to become $D[12:0]$, and then the bits $D[16:13]$ are padded with zeros to form the input signal $D[16:0]$ for the shift adder. The other input, $R[16:4]$, is initialized to 0 via DFF. These two input signals are then added using the MCSA-17b shift adder. The timing diagram of the MCPA is shown in **Fig. 8(b)**.

In the first cycle, the result $R\,[20:4]$ contains $R\,[20:17] = 0$, so these bits do not participate in the computation in the second cycle. The $R\,[16:4]$, which correspond to $Sum_1[16:4]$, are returned as input for the second cycle to the MCSA-17b.

In the second cycle, $Sum_2[16:0]$ is directly input as $D[16:0]$ into the MCSA-17b to be added with $R[16:4]$ after shifting. The result $R[20:4]$ combined with $R_1[3:0]$ gives the final sum result $FS\,[20:0]$.

$$FS\,[20:0] = \sum_{i=0}^{31} IN_i\,[7:0] \times W_i\,[7:0] \qquad (11)$$

Conventional post-adders require a 21-bit adder (composed of 20 full adders and 1 half adder) to complete this process, whereas the MCSA-17b requires only 12 full adders and 5 half adders. The power consumption comparison between the two will be discussed in **Section IV**

G. Macro Structure and Data Flow

As shown in **Fig. 9**, the 32 × 32 × 128-bit MRAM computation array is divided into 32 MRAM sub-arrays, with each sub-array containing two Booth Computing Blocks. Each BCB consists of 16 × 128-bit 2T1M cells, 128 VDRs, and 32 Booth computing groups (BCG). Each BCG contains two Booth encoders, one Booth decoder, and one BCA. The system can operate in either memory mode or compute mode, supporting 4b-IN/4b-W/13b-OUT and 8b-IN/8b-W/21b-OUT. The computation and flow of data within the BaM-CIM are illustrated in **Fig. 10**.

Taking the MAC operation of $IN_{0-31}\,[7:0]$ = 01110101(117) and $W_{0-31}\,[7:0]$ = 00100011(35) as an example: In the first cycle, $IN_{0-31}\,[3:0]$ = 0101(5) enters the Booth Encoder, yielding the control signals NEGB=0, TWOB=0, ZEROB=0; NEGC=0, TWOC=0, ZEROC=0. The control signals and the weight W are processed by the Booth Decoder to generate the partial products A, B, and C.

Multiplying $IN_0\,[3:0]$ with $W_0\,[3:0]$:The partial products obtained are $A\,[7:0]$ = 00000000, $B\,[7:0]$ = 00001100$C\,[7:0]$ = 00000011. These three partial products are summed in the BCA, resulting in $WAS_0\,[7:0]$ = 00001111(15). A total of 32 sets of $WAS\,[7:0]$ are fed into the Adder Tree for summation, yielding $Sum_L\,[12:0]$ = 0000111100000(480).

Multiplying $IN_0\,[3:0]$ with $W_0\,[7:4]$:The partial products obtained are $A\,[7:0]$ = 00000000, $B\,[7:0]$ = 00001000, $C\,[7:0]$ = 00000010. These three partial products are summed in the BCA, resulting in $WAS_0\,[7:0]$ = 00001010(10). A total of 32 sets of $WAS\,[7:0]$ are fed into the Adder Tree for summation, yielding $Sum_H\,[12:0]$ = 0000101000000(320).

$Sum_L\,[12:0]$ and $Sum_H\,[12:0]$ are then processed by the MCSA-13b for shift-and-add summation, producing $Sum_1\,[16:0]$ = 00001010111100000(5600).

Similarly, in the second cycle, $IN_{0-31}\,[7:4]$ = 0111(7) enters the Booth Encoder and is computed with $W_{0-31}\,[7:0]$, resulting in $Sum_2\,[16:0]$ = 00001111010100000(7840). $Sum_1\,[16:0]$ and $Sum_2\,[16:0]$ are then combined through a shift-and-add operation in the MCPA to produce the final result $FS\,[20:0]$.

H. Data Mapping and CNN Evaluation

BaM-CIM is designed for CNNs with a large number of MAC operations. **Fig. 11** illustrates the bitwise mapping scheme, where the 8-bit weights of 32 channels are stored in the BaM-CIM. BCB_0 stores the lower four bits of the weights, while BCB_1 stores the upper four bits. The input features of the 32 channels are fed into the BaM-CIM over two cycles and undergo MAC operations with the corresponding weights.

An 8-bit quantized CNN using the ResNet-18 and ResNet-50 architectures was implemented with the PyTorch package to validate the accuracy of the proposed BaM-CIM. On the CIFAR-10 dataset, the accuracies achieved with ResNet-18 and ResNet-50 were 95.31% and 95.24%, respectively. On the CIFAR-100 dataset, ResNet-18 achieved a Top-1 accuracy of 75.81% and a Top-5 accuracy of 92.76%, while ResNet-50 achieved a Top-1 accuracy of 76.24% and a Top-5 accuracy of 93.08%, as shown in **Table III**.

IV. Simulation Results and Analysis

The proposed BaM-CIM architecture was simulated in Cadence Virtuoso using a hybrid technology of VGSOT-MTJ and GAA-CNTFET, modeled with Verilog-A, as shown in **Table IV**. Key parameters of GAA-CNTFET and VGSOT-MTJ are listed in **Table V** [43]. In this study, 200% TMR was selected to achieve the proposed circuit design. The minimum gate width of the GAA-CNTFETs in the proposed CIM macro was 18 nm, and the V_{fb} of CNTFETs was illustrated in the schematic. The memory size is 32 × 32 × 128 to obtain a 128 kb array.

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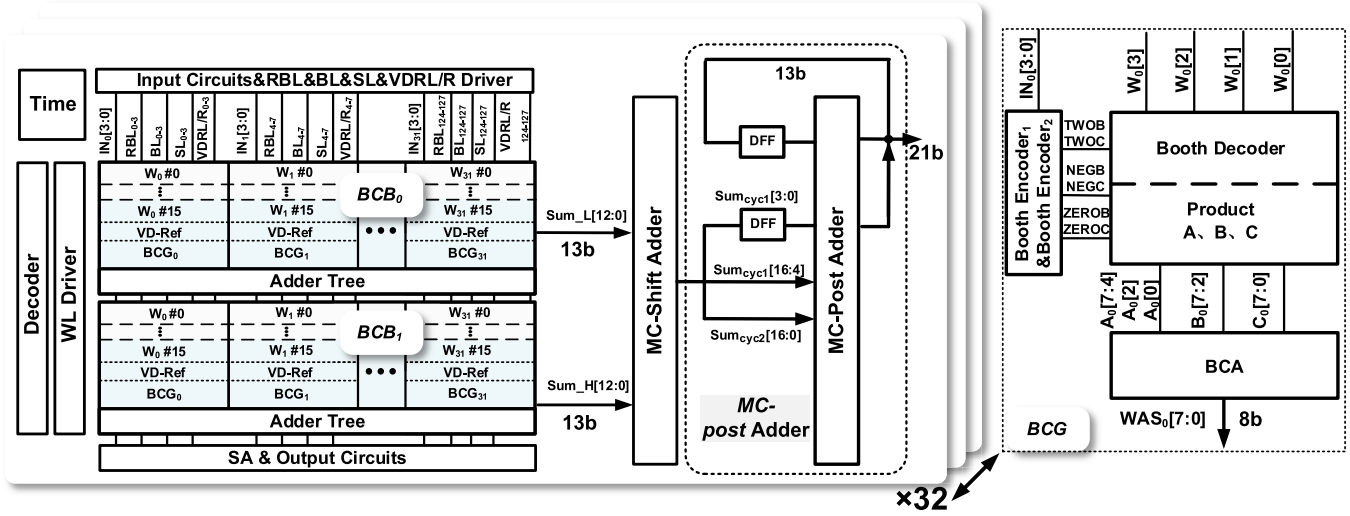


Fig. 9. Overall structure of the BaM-CIM architecture.

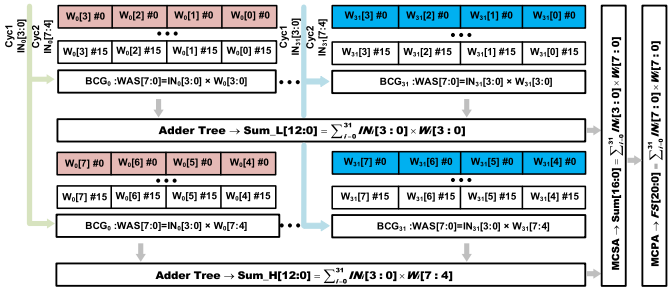


Fig. 10. Data flow of BaM-CIM.

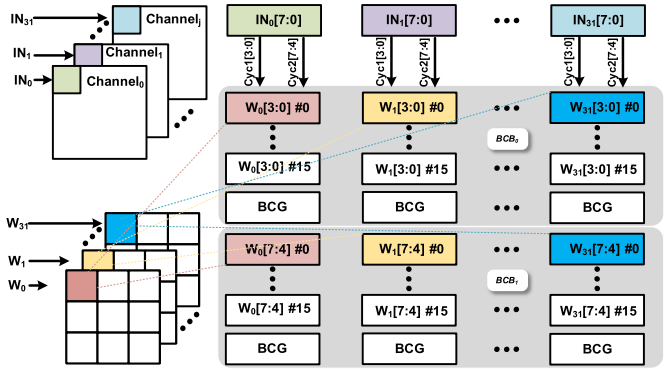


Fig. 11. Weight mapping of 32 channels MAC in BaM-CIM.

TABLE III
NN DEMONSTRATIONS

Dataset	CIFAR-10	CIFAR-10	CIFAR-100	CIFAR-100	CIFAR-100
NN model	ResNet-18	ResNet-50	ResNet-18	ResNet-50	ResNet-50
Bit Precision	8b/8b IN/W	8b/8b IN/W	8b/8b IN/W	8b/8b IN/W	FP
Accuracy	95.31%	95.24%	75.81%	76.24%	77.12%

A. Robustness of the VDR

To analyze the stability of the data read operation, we evaluated the V_M voltage and sensing margin for different TMR values under a 0.8V supply voltage, as shown in **Fig. 12(a)**. The sensing margin is defined as $(V_{MH} - V_{ML})$.

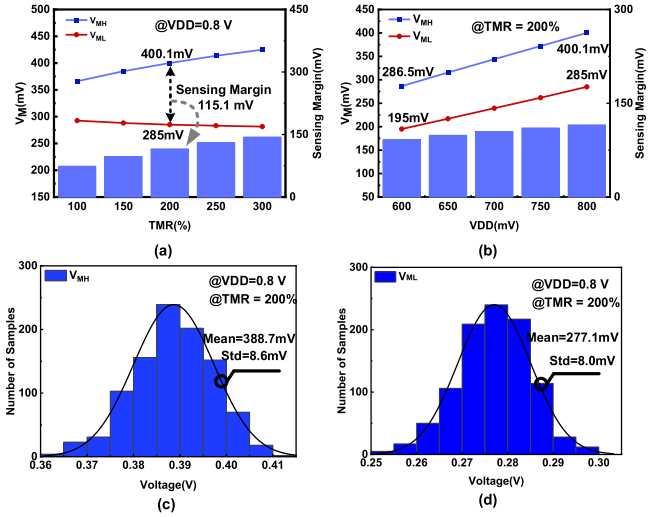


Fig. 12. (a) Simulation results of the V_M and Sensing Margin with different TMR. (b) Simulation results of the V_M and Sensing Margin with different VDD. (c) Monte Carlo simulations of V_{MH}. (d) Monte Carlo simulations of V_{ML}.

TABLE IV
SIMULATION TOOL AND MODEL

Simulation Tool	Cadence Virtuoso
Model	12nm GAA-CNTFET VGSOT-MTJ
Model Source (CNTFET)	https://nano.stanford.edu/downloads/vs-cnfet-model
Model Source (VGSOT-MTJ)	http://www.spinlib.com/VGSOT_MTJ.html

A higher TMR for the VGSOT-MTJ results in an increased voltage difference between V_{MH} and V_{ML}, leading to a higher reading margin. When the TMR is 100%, V_{MH} and V_{ML} are 366 mV and 292 mV, respectively, yielding a sensing margin of 73.5 mV. When the TMR reaches 200%, V_{MH} and V_{ML} are 400 mV and 285 mV, respectively, providing a sensing margin of 115 mV. This sensing margin can be easily distinguished by an LVT inverter. When the TMR reaches 300%, a sensing margin of 143.5 mV is obtained. In this

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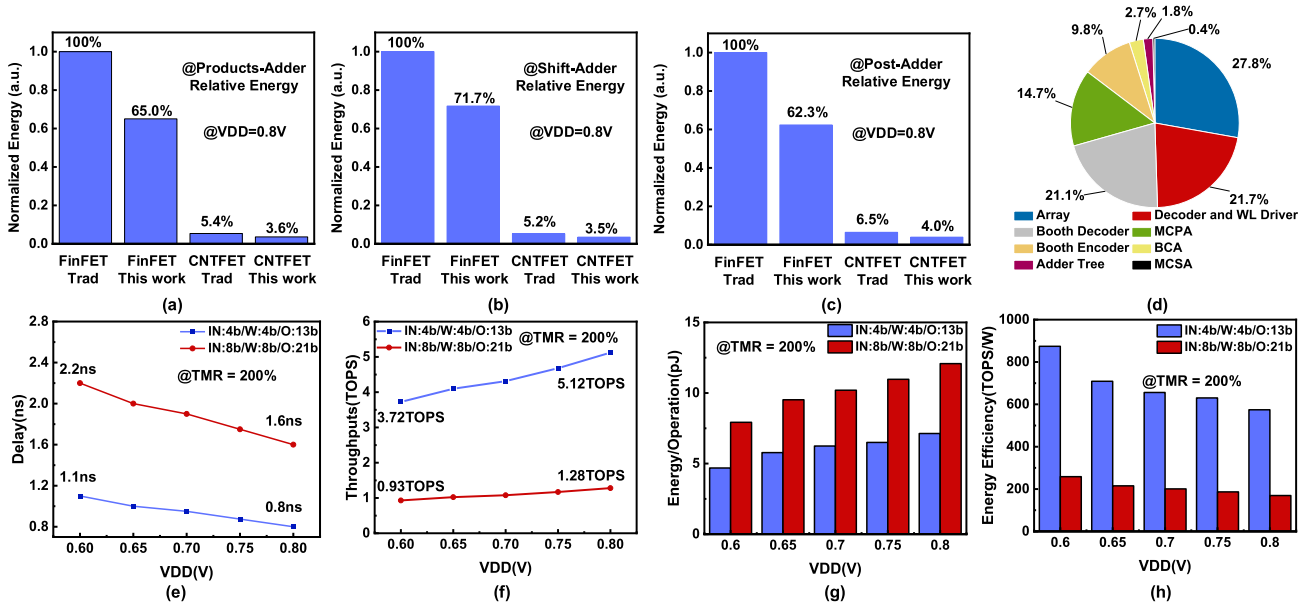


Fig. 13. (a) Energy consumption of different Products-Adders implemented using 14nm-FinFET and GAA-CNTFET. (b) Energy consumption of different Shift-Adders implemented using 14nm-FinFET and GAA-CNTFET. (c) Energy consumption of different Post-Adders implemented using 14nm-FinFET and GAA-CNTFET. (d) Energy consumption distribution of BaM-CIM. (e) Simulation results of the delay. (f) Simulation results of throughput. (g) Simulation results of the energy consumption. (h) Simulation results of energy efficiency.

TABLE V
IMPORTANT PARAMETERS OF GAA-CNTFET AND VGSOT-MTJ MODEL

	Parameters	Value
GAA-CNTFET	Transistor width (W)	18nm
	Physical gate length (Lg)	12nm
	Contact length (Lc)	12nm
	Source/drain extension length (Lext)	3nm
	CNT diameter (d)	1.2nm
	Gate oxide thickness (tox)	3nm
	Gate height (Hg)	20nm
	Flat band voltage (V_{fb})	0.16/-0.315 V
	Spacing between the CNTs (s)	3nm
VGSOT-MTJ	Diameter of MTJ (D)	50nm
	MgO barrier thickness (tox)	1.4nm
	TMR ratio under zero bias voltage (TMR_0)	200%
	MTJ surface length (a)	50nm
	MTJ surface width (b)	50nm
	MTJ surface radius (r)	25nm
	AFM-strip width (W)	50nm
	AFM-strip thickness (d)	3nm
	AFM-strip length (l)	60nm

work, a 200% TMR was selected to implement the proposed CIM design. Similarly, the values of V_{MH} and V_{ML} also vary under different VDD levels, as shown in **Fig. 12(b)**. When VDD is 600 mV, V_{MH} and V_{ML} are 286.5 mV and 195 mV, respectively, with a sensing margin of 91.5 mV. When VDD is increased to 800 mV, V_{MH} and V_{ML} increase to 400.1 mV and 285 mV, respectively, and the sensing margin increases to 115 mV.

To further evaluate the robustness of the data read operation, we conducted a 1k Monte Carlo simulation of the V_M under VDD = 0.8V and TMR = 200%, as shown in Figs. 12(c), (d). The V_M can exist in two states, V_{MH} and V_{ML}, depending on the MTJ resistance. For the Monte Carlo simulation, important

TABLE VI
MONTE CARLO PARAMETERS OF GAA-CNTFET AND VGSOT-MTJ MODEL

	Monte Carlo Parameters	±3δ Variation
GAA-CNTFET	Physical gate length (Lg)	10%
	Contact length (Lc)	10%
	Gate height (Hg)	10%
	Gate oxide thickness (tox)	10%
	Flat band voltage (V_{fb})	10%
	The number of CNTFET (Ncnt)	10%
MTJ	Resistance-area (RA)	10%
	Free layer thickness (tsl)	10%
	TMR ratio under zero bias voltage (TMR_0)	10%

process parameters for the VGSOT-MTJ were considered with a 10% variation [33]. Additionally, the key process parameters for the GAA-CNTFETs were considered with a 3σ variation of 10% [44], where Ncnt is calculated based on the transistor width and the spacing (s) between the carbon nanotubes, as shown in **Table VI**. When the MTJ is in a low-resistance state, the mean of the V_{MH} is 388.7 mV, with a standard deviation of 8.6 mV. When the MTJ is in a high-resistance state, the mean of the V_{ML} is 277.1 mV, with a standard deviation of 8.0 mV. The sensing margin remains around 111.6 mV. In this case, 1k Monte Carlo simulations were performed on the output of the LVT inverter, and the LVT inverter correctly output results in all cases without any errors.

B. The Performance Analysis of Adders

The energy consumption of four types of adders, including the traditional adder built with 14nm-FinFET, the improved adder built with 14nm-FinFET, the traditional adder built with

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GAA-CNTFET, and the improved adder built with GAA-CNTFET, was compared. The energy consumption of these adders was normalized, as shown in **Figs. 13(a), (b)**, and **(c)**. **Fig. 13(a)** shows the energy consumption comparison of the partial product adders (BCA and traditional scheme). The adder built with 14nm-FinFET consumes significantly more energy than the GAA-CNTFET-based adder group, and the BCA saves approximately 35% in energy consumption compared to the traditional scheme when using the same transistor technology. **Fig. 13(b)** shows the energy consumption comparison of the shift product adders (MCSA and traditional scheme). Again, the 14nm-FinFET-based group has higher energy consumption, with MCSA saving about 30% compared to the traditional scheme. **Fig. 13(c)** presents the energy consumption comparison of the post-adder (MCPA and traditional scheme), where MCPA saves around 38% in energy consumption compared to the traditional scheme. These results demonstrate that the optimization of the three adders plays a positive role in enhancing the overall circuit performance.

C. The Performance Analysis of BAM-CIM

The energy consumption distribution is shown in **Fig. 13(d)**. The array accounts for the largest portion of the total energy consumption, at 27.8%. Among the peripheral computing circuits, the Booth Decoder consumes the most energy, at 21.1%, while the Booth Encoder accounts for 9.8%. The energy consumption of the BCA is 2.7%, while the MCSA and MCPA account for 0.4% and 14.7%, respectively.

Fig. 13(e) shows the computation delay of the proposed design in CIM mode under different VDD levels. When VDD is 0.8 V, the Booth operation, local addition, shifting, global addition, and post-addition operations can be completed within 0.8 ns. Therefore, when VDD is 0.8 V and TMR is 200%, the computation delays for 4b-IN/4b-W and 8b-IN/8b-W are 0.8 ns and 1.6 ns, respectively. In this case, the entire circuit is activated to perform 64 sets of 4b-W × 4b-IN with 128 MAC operations or 32 sets of 8b-W × 8b-IN with 64 MAC operations.

Fig. 13(f) shows the throughput of the circuit as a function of VDD. As the supply voltage increases, the operation delay of the circuit decreases, leading to a higher throughput. At VDD = 0.8 V, the throughput is 5.12 TOPS (64 × 2 × 32 OP / 0.8 ns) for 4b-IN/4b-W and 1.28 TOPS (32 × 2 × 32 OP / 1.6 ns) for 8b-IN/8b-W.

Fig. 13(g) shows the energy consumption of the proposed design in CIM mode under different VDD levels. As the supply voltage increases, the energy consumption of the MAC operation also increases. The energy consumption of the MAC operation with 4b-IN/4b-W is 4.7 pJ/MAC at VDD = 0.6 V and 7.1 pJ/MAC at VDD = 0.8 V. The energy consumption of the MAC operation with 8b-IN/8b-W is 7.9 pJ/MAC at VDD = 0.6 V and 12.1 pJ/MAC at VDD = 0.8 V. The MAC with 8b-IN/8b-W/21b-OUT can complete the operation within 1.6 ns (2 cycles).

Fig. 13(h) presents the energy efficiency results of the CIM operation under TMR = 200% and different VDD levels. Although throughput improves with higher voltage, the circuit energy consumption also increases, resulting in a decrease in energy efficiency. For 4b-IN/4b-W, at VDD = 0.6 V and VDD = 0.8 V, the energy efficiency of the MAC operation is 874.1 (64 × 2 × 32 OP / 4.7 pJ) and 574.3 (64 × 2 × 32 OP / 7.1 pJ) TOPS/W, respectively. Similarly, for 8b-IN/8b-W, at VDD = 0.6 V and VDD = 0.8 V, the energy efficiency is 258.4 (32 × 2 × 32 OP / 7.9 pJ) and 169.5 (32 × 2 × 32 OP / 12.1 pJ) TOPS/W, respectively.

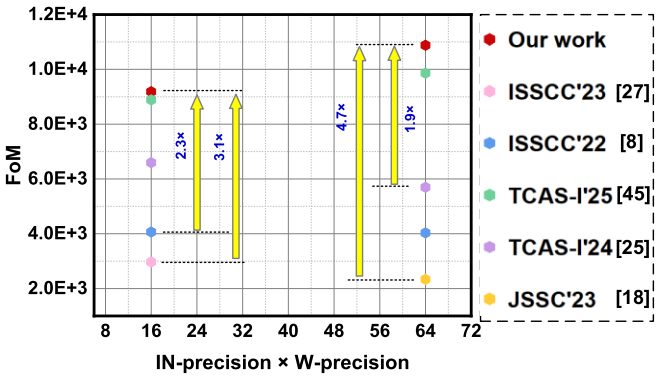


Fig. 14. FoM comparison with state-of-the-art CIM macros.

D. Comparison With Other Works

Table VII presents a comprehensive performance comparison between BaM-CIM and state-of-the-art CIM architectures. From the computational unit perspective, BaM-CIM maintains the fundamental 2T1M cell configuration using SOT-MTJ devices without additional components, thereby enhancing area efficiency.

Regarding operational cycles, the Booth algorithm implementation enables BaM-CIM to complete 8b-IN/8b-W MAC operations within two clock cycles. Benefiting from the high I_{ON}/I_{OFF} ratio characteristics of CNTFETs, each computational cycle requires merely 0.8 ns, granting BaM-CIM superior operational speed compared to most CIM implementations.

While [18] and [28] similarly employ Booth algorithms for computational acceleration, [18] exhibits suboptimal energy efficiency due to an unoptimized adder design. [28] adopts a sequential partial product generation approach (one partial product per cycle) rather than the concurrent generation scheme proposed in this work, resulting in extended computation cycles. Furthermore, its CMOS-based implementation demonstrates inferior switching speed compared to CNT-FET technology, thereby prolonging computation time.

Despite [25] and [45] employing identical devices to this work, it suffers from significantly increased computation cycles and energy consumption due to its bit-serial computation strategy. [19] Implement analog CIM architectures that achieve higher computational parallelism at the expense of reduced precision in computation accuracy.

The figure of merit (FoM), defined as energy efficiency× IN-precision × W-precision × out ratio, is employed to compare the performance among various CIM architectures. Benefiting from the Booth algorithm and the optimized adders, the proposed BaM-CIM achieves FoM values of 9,184 (calculated as 574 TOPS/W × 4 × 4 × 1) and 10,848 (calculated

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TABLE VII
COMPARISON OF BAM-CIM WITH THE PREVIOUS STUDIES

	This work	ISSCC'23 [27]	ISSCC'22 [8]	TCAS-I'25 [45]	TCAS-I'24 [25]	TCAS-I'21 [23]	JSSC'23 [18]	TNANO'24 [28]	TCAS-I'24 [19]
Technology	**12nm GAA-CNTFET /VGSOT-MTJ**	22nm CMOS/STT-MTJ	5nm CMOS	12nm GAA-CNTFET /VGSOT-MTJ	12nm GAA-CNTFET /VGSOT-MTJ	28nm CMOS /SOT-MTJ	28nm CMOS	40nm CMOS /SOT-MTJ	14nm CMOS /STT-MTJ
Array Size	**128Kb**	8Mb	64Kb	288Kb	288Kb	1MB	12KB	NA	2Kb
Cell Type	**2T1M**	1T1M	12T	4T2M	4T2M	2T1M	6T	2T1M	3T1M
Supply Voltage (V)	**0.6-0.8**	0.65-0.8	0.5-0.9	0.6-0.8	0.6-0.8	1	0.6-1.0	1	0.8
Access Time[5] (ns)	**0.8** [3,7] **1.6** [3,8]	20	2.78 [7] 5.56 [8]	1.5[3,7] 3.1[3,8]	4 [3,7] 8 [3,8]	253.58	NA	16.6	36
Cycle Numbers[6]	**1** [7] **2** [8]	8	4 [7] 8 [8]	2 [7] 4 [8]	4 [7] 8 [8]	8	NA	4	8
Input/Weight/ Output Precision	**4b/4b/13b**	8b/8b/26b	4b/4b/14b	4b/4b/16b	4b/4b/12b	4b/4b/11b	8b/8b/NA (INT Mode)	8b/8b/NA	8b/8b/20b
	8b/8b/21b		8b/8b/22b	8b/8b/24b	8b/8b/20b				
Verification	**Simulation**	Fabricated	Fabricated	Simulation	Simulation	Simulation	Fabricated	Simulation	Simulation
Operation Mode	**Digital CIM**	Digital CIM	Digital CIM	Digital CIM	Digital CIM	Digital CIM	Digital CIM	Digital CIM	Analog CIM
Computing Method	**Booth Algorithm**	Bit-Serial	Bit-Serial	Bit-Serial	Bit-Serial	Bit-Serial	Booth Algorithm	Booth Algorithm	Bit-Serial
Throughput (TOPS)	**3.72**[1,4,7]**-5.12** [1,3,7]	0.59[2,8]	2.94 [2,7]	8.78[1,4,7]-12.29 [1,3,7]	4.61 [1,3,7]	NA	1.35 [2,8]	NA	NA
	0.93[1,4,8]**-1.28** [1,3,8]		0.73 [2,8]	2.25[1,4,8]-2.97 [1,3,8]	1.15 [1,3,8]				
Energy Efficiency (TOPS/W)	**574**[1,3,7]**- 874** [1,4,7]	46.4 [2,8] - 160.1 [2,8]	254 [2,7]	555[1,3,7]- 830 [1,4,7]	134.6 [1,3,7]	12.28 [1]	36.5 [2,8]	12.4 [1]	79.07 [1]
	170[1,3,8]**- 258** [1,4,8]		63 [2,8]	156[1,3,8]- 226 [1,4,8]	33.7 [1,3,8]				

[1]Simulation; [2]Chip measurement; [3]Performance @VDD=0.8 V; [4]Performance @VDD=0.6 V; [5]The time required to complete one CIM operation; [6]The number of calculation cycles; [7]Performance @ 4b-IN/4b-W; [8]Performance @ 8b-IN/8b-W;

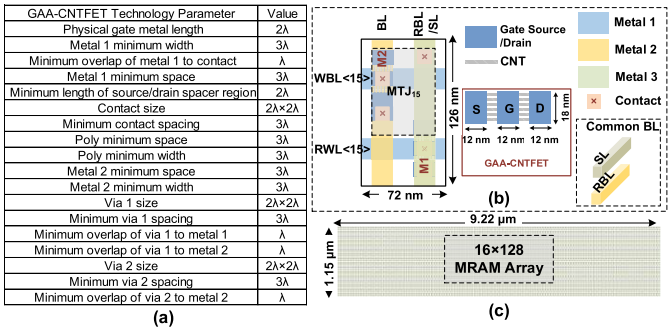


Fig. 15. (a) Layout design rules of the 12 nm GAA-CNTFET technology node. The minimum feature size λ = 6 nm. (b) The layout of 2T1M cell. (c) The layout of 16 × 128 12-nm GAA-CNTFET MRAM mini-array.

as 169.5 TOPS/W × 8 × 8 × 1) at 4b-IN/4b-W/16b-OUT and 8b-IN/8b-W/24b-OUT, respectively, as shown in **Fig. 14**. At 4b-IN/4b-W, the FoM of BaM-CIM is 2.3× and 3.1× higher than [8] and [27], respectively, while at 8b-IN/8b-W, it is 1.9× and 4.7× higher than [18] and [25], respectively. BaM-CIM exhibits comparable FoM to [45] but outperforms it at both 4b-IN/4b-W and 8b-IN/8b-W.

E. Layout Evaluation

Since neither VGSOT-MTJ nor GAA-CNTFET is commercially available, and corresponding commercial process libraries are lacking, we designed the layout of the proposed circuit based on λ-based layout design rules for 12 nm GAA-CNTFET technology, as shown in **Fig. 15(a)**. Here, λ (6 nm in this work) represents the minimum feature size, which is considered half of the physical gate length [35], [36], [46], [47].

The layout of the 2T1M cell is shown in **Fig. 15(b)**. Each cell includes BL, RBL, and SL. By applying the common area (CA) technique, RBL and SL are positioned simultaneously in different metal layers. BL and RBL are located in metal-2, while SL is placed in metal-3, thereby reducing the width of the 2T1M cell to 12λ and further minimizing its area. **Fig. 15(c)** presents the layout of a 16 × 128 12nm GAA-CNTFET MRAM mini-array with an area of 10.603 μm^2.

Owing to the back-end-of-the-line-compatible process compatibility between SOT-MRAM and CNTFET, a three-dimensional (3D) vertical stacking architecture can be employed to further enhance storage density: a layer of CNTFETs is designed and fabricated above the first layer of SOT-MRAM/CNTFET, insulated by a SiO2 interlayer. On this basis, a second layer of MTJs is fabricated by repeating the manufacturing process. Interlayer connections are realized through through-silicon vias (TSVs), with copper serving as the conductive filling material. This approach effectively improves area efficiency [45].

V. CONCLUSION

This work proposes a Booth algorithm-based CIM architecture leveraging VGSOT-MTJ and GAA-CNTFET technologies to achieve energy-efficient MAC operations. Firstly, a novel VDR computational cell is introduced, utilizing a 2T1M cell structure with reference-unit-based resistive voltage division to achieve 115 mV read margin at 0.8 V supply voltage, while demonstrating scalability through adjustable operating voltages and TMR ratios. Secondly, the implementation of modified Booth encoding reduces partial product quantities by 25% for both 4b-IN/4b-W and 8b-IN/8b-W MAC operations, enabling 8b-IN/8b-W computation within two 0.8ns cycles through CNTFETs' superior switching characteristics. Thirdly, BCA is designed by leveraging the special characteristics of partial products. Compared with traditional adders, the energy consumption of BCA can be reduced by 35%. Moreover,

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the subsequent adders are improved to MCSA and MCPA. Compared with traditional shift adder and post adder, the energy consumption of MCSA and MCPA can be reduced by 30% and 38% respectively. Finally, the performance of a 128kb BaM-CIM system is analyzed. At an operating voltage of 0.8V, for 4b-IN/4b-W MAC operations, the throughput reaches 5.12 TOPS and the energy efficiency is 574.3 TOPS/W. For 8b-IN/8b-W MAC operations, the throughput is 1.28 TOPS and the energy efficiency is 169.5 TOPS/W. Compared with other similar CIM architectures, it has certain advantages in terms of energy efficiency and throughput. Therefore, this work provides a solution for high-efficiency MAC operations.

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